

SUMMARY

Innovative Software Developer & Research Scholar with 4+ years of experience in application development, Machine learning, deep learning, image processing and IoT solutions. Skilled in designing and implementing intelligent systems for health monitoring, image processing, and real-time analytics. Experienced in both academic research and industry projects, with writing experience of publications, patents. and cross functional collaboration with researchers and developers.

WORK EXPERIENCE

Research Assistant (MS – Research Scholar) | Indian Institute of Technology, Palakkad | August 2024 – Present

- Led research on resource-efficient road boundary determination using Image processing techniques- Conference paper accepted
- Lane crack or pothole detection - image enhancement and adaptive thresholding for robust detection - Preference journal papers

Junior Research Fellow(JRF) | Indian Institute of Technology, Palakkad | Jun 2023 - July 2024

- Human Activity and Heart Rate Monitoring (Patent documented): Android application integrating GPS and accelerometer for human acitvity aware,real-time pulse rate tracking using PPG sensor board.
- Real-Time Bluetooth-Based ECG/PPG Monitoring: Android application for dynamic visualization of biomedical signal waveforms using ECG bluetooth sensor board.

Python Junior Odoo Developer | Loyal IT Solutions, Infopark Koratty, Thrissur | Dec 2023 – Jun

- Developed and customized Odoo ERP modules integrating CRM, sales, project management, and accounting functionalities.
- Improved ERP efficiency by implementing automated workflows, optimized front-end and back-end processes

Python Odoo Developer Trainee | Cybrosys Technologies Pvt Ltd, Kinfra Techno Park, Calicut | August 2022 - 6 Months Trainee

- Designed and developed business management applications for diverse client requirements.
- Enhanced existing ERP systems by debugging, testing, and adding new features.

CORE TECHNICAL SKILLS

- Machine learning, Deep learning, Image processing
- Programming & Development: Python, Java, MATLAB, Kotlin, Golang, PHP, C, React.js, Angular, HTML/CSS, Odoo JS, XML, PSQL
- Mobile & Web Applications: Android Studio (Java/XML), Odoo ERP, GUI tools, Figma
- Machine & Deep Learning: CNN (TensorFlow), YOLO, OpenCV, Computer Vision
- Databases & Tools: MySQL, PostgreSQL, MS Office (Word, Excel, PowerPoint)
- Other Expertise: Image Editing, Video Editing, Poster Design, Content Creation

EDUCATION

- M.S. in Electrical Engineering – Indian Institute of Technology, Palakkad (2024–2026) – 82.3%
- B.Tech in Computer Science – Royal College of Engineering & Technology, Thrissur -(2018–2022) – 81%
- Higher Secondary (Science) – MES HSS Ponnani (2016–2018) – 84%
- CBSE Class 10 – Amal English School, Thrissur (2016) – 86%

ACHIEVEMENTS & LEADERSHIP

- MS Representative - Research Affairs Council – IIT Palakkad - 2025-26
 - MS Overall Placement Coordinator – IIT Palakkad
 - Champion in Throwball & Kabaddi; Winner in Shot Put
 - Awards in Debate, Elocution
 - Basketball, Badminton, Volleyball, Throwball
 - Motivational Speaker & Event Organizer for academic and cultural events
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MAIN RESEARCH AREAS

- Signal Processing – Biomedical signal analysis, ECG/PPG monitoring, human personal activity (sitting, walking, running) monitoring using gyroscope sensor, human activity recognition.
 - Image Processing & Computer Vision – Road boundary detection, pothole detection, lane detection, crop color-based disease and solution recognition
 - Machine Learning & Deep Learning – Lightweight models for edge devices (smartphones, drones) CNN, YOLO, sequence-to-sequence models (Transformer based language models)
 - IoT & Embedded Systems – IoT-GPS enabled health monitoring, FPGA/Raspberry Pi based deployment - planning works.
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PUBLICATIONS

- Deep Learning for Mask Detection and Sanitise Checker in Public during COVID-19, IJIRCCE, 2020.
 - Automatic Text Generation Using Natural language processing, IJIRCCE, 2021.
 - Vision-Based Pixel-Level Road Boundary Segmentation Under Low-Light Night-Time Driving Conditions - accepted - Organised by Bits Pilani, Dubai, Campus - April 2026
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PROJECT HIGHLIGHTS - IIT PALAKKAD

- Real-Time Road Boundary Detection Using Deep Learning-Based Segmentation with Raspberry Pi Camera Interfacing for Night-Time Dashboard Applications—an Embedded System—MS Thesis.
- FPGA-accelerated deployment and evaluation of a segmentation model on the PYNQ-Z2 platform.
- Lane Determination—Applied image processing techniques and other deep learning techniques for lane boundary for night challenging.
- Human Activity-Aware Heart Rate Monitoring (Patent Pending) – Android app integrating GPS and accelerometer for real-time pulse tracking.
- Real-Time Bluetooth ECG/PPG Monitoring – Android app for dynamic waveform visualization of biomedical signals.
- Tomato Leaf Disease Detection – Image processing + deep learning for early disease classification – Course project.
- Deep Learning-based Emotion Recognition – Integrated for face detection + RepVGG for real-time facial emotion classification – Course project.
- Text-to-Speech in the Malayalam Language—TensorFlow-based sequence-to-sequence model with Transformer + HiFi-GAN/WaveGlow vocoders.

TUTOR EXPERIENCE

- Assisted undergraduate students in understanding core concepts in engineering subjects.
 - Conducted problem-solving sessions and clarified doubts to improve students' academic performance in programming language and Data analytics organized by IPTIF, IIT Palakkad, Kerala, India.
 - Worked as a tutor - session include Data analytics in subject with Data Science through Online session and remote.
 - Simplified complex topics through clear explanations and practical examples.
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WORKSHOP ASSISTANCE

- Assisted in a Python and Data Analytics Workshop (2026), supporting participants with programming exercises and basic data analysis tasks, organized by IPTIF, IIT Palakkad.
 - Assisted in a VLSI Workshop (April 2024), helping with technical coordination and participant guidance and students from various part of Kerala.
 - Contributed to the Pre-Research Scholar Day Workshop on Python Programming and Machine Learning by assisting in session organization and hands-on activities.
 - Supported instructors and participants during practical sessions and technical discussions.
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WORKSHOP EXPERIENCE

- Participated in academic workshops related to machine learning and data analysis.
 - Gained hands-on experience with tools and techniques used in research and data processing.
 - Collaborated with peers to complete practical exercises and technical discussions.
 - Applied workshop knowledge to research and project development.
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